

Shubham Srivastava

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Summary

AI Engineer and Data Science enthusiast specializing in computer vision, deep learning, NLP, and machine learning deployment. Experienced in developing end-to-end AI systems involving model training, evaluation, inference APIs, and real-time monitoring solutions using Python, TensorFlow, OpenCV, and backend technologies.

Education

B.Tech in Computer Science and Engineering (Data Science)

Haridwar University, Roorkee, India
CGPA: 8/10 (till 6th semester)

Sept 2023 – June 2027

Experience

AI/ML Intern

CoDA Lab, IIT Roorkee

On-site

May 2026 – July 2026

- Developed Aqua Vision, a hybrid computer vision framework integrating 3 deep learning models with an ESP32-CAM edge device for automated water pollution monitoring.
- Unified 5+ application modules covering inference, testing, and deployment into a unified environmental monitoring system.
- Contributed to backend inference server development, real-time monitoring dashboard integration for 2 zones, system testing, and deployment workflow for an end-to-end AI monitoring application.

Industrial AI Intern

Aditya Birla Group (Hindalco)

On-site

July 2025 – Aug 2025

- Engineered a real-time PPE monitoring system using YOLOv8 and DeepSORT for industrial safety
- Implemented automated alert workflows for real-time PPE compliance monitoring across 5 safety equipment classes
- Designed sensor-data analytics pipelines for predictive maintenance insights, contributing towards approximately 20% reduction in machine downtime.

AI/ML Intern

Microsoft TechSaksham

Remote

Dec 2024 – Feb 2025

- Trained a CNN-based plant disease classification model achieving 95% accuracy on a dataset of 5,000+ images
- Optimized preprocessing and inference workflows, reducing false prediction rates by 40%.

Technical Skills

Languages: Python, Java, SQL

Machine Learning: TensorFlow, Keras, Pandas, NumPy

Computer Vision: OpenCV, YOLO, Object Detection, Image Classification, Image Processing

NLP & GenAI: Transformers, Sentence Transformers, Embeddings, NER, Semantic Search, RAG

Backend & Deployment: Flask, REST APIs, Docker, AWS EC2, Git, React

Databases: MongoDB, MySQL

Projects

1. Aqua Vision: Hybrid Framework for Pollution Detection and Quality Assessment (CV) [GitHub](#) | [Demo](#)

- Implemented a YOLO-based waste detection pipeline achieving 88.26% mAP@50 for pollution object detection.
- Built deep learning models for turbidity estimation and water quality classification achieving 94.46% and 95.78% accuracy respectively.

2. Persona-Driven Document Analyst (RAG/NLP) [GitHub](#)

- Created a semantic document intelligence pipeline using sentence transformers, embeddings, and retrieval-based NLP techniques.
- Designed document understanding workflows combining semantic search and Named Entity Recognition, achieving 92% precision for entity extraction.

3. Fraud Anomaly Detection System (DBSCAN) [GitHub](#) | [Demo](#)

- Constructed an unsupervised fraud detection pipeline using DBSCAN clustering over 200K+ transaction records.
- Achieved a 0.62 silhouette score for anomaly clustering evaluation and reduced false positives by 25% through improved feature analysis.

Research Publication

“Inclusive Voice-AI for Rural India: A Deep Learning Framework for Dialectal and Low-Resource Adaptation.”
International Journal for Multidisciplinary Research (IJFMR), Volume 8, Issue 1 (2026)

Achievements & Certifications

- Runner-Up** - MDG Space Hackathon, IIT Roorkee, 2023
- Second Runner-Up** - Smart India Hackathon (Internal), 2024
- Second Runner-Up** - Project Exhibition (Group A), Haridwar University, 2026
- Chairperson** - Trojan Club, 2025 to Present
- OCI 2025 Certified Data Science Professional** — Oracle
- Machine Learning with Python** — IBM Coursera
- Python for Data Science** — NPTEL, IIT Madras